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Bioinputs as a Sustainable Strategy in Rice and Bean Production: Implications for Health and Planetary Nutrition

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Citation
WILHAN SANTOS, ANA PAULA SIQUEIRA, RITA DE CÁSSIA NOGUEIRA DE MOURA, MARIA LUIZA FERREIRA DE SOUSA E SILVA. Bioinputs as a Sustainable Strategy in Rice and Bean Production: Implications for Health and Planetary Nutrition. This is a training record and not for publication.

REVIEW TITLE AND BASIC DETAILS

Review title

Bioinputs as a Sustainable Strategy in Rice and Bean Production: Implications for Health and Planetary Nutrition

Condition or domain being studied

Biological Treatment

This review focuses on sustainable agricultural systems, specifically rice (Oryza sativa) and common bean (Phaseolus vulgaris) production, which are key staples for food security, particularly in Brazil. The domain of interest involves the application of bioinputs (e.g., biofertilizers, biopesticides, and biostimulants) as alternatives to conventional synthetic inputs. The review explores how these biological technologies influence agronomic performance, soil health, environmental sustainability, and their broader implications for food systems and planetary health. This approach aligns with the growing need to integrate agricultural productivity with environmental conservation and human health outcomes.

Rationale for the review

The growing global demand for sustainable food systems has intensified the search for agricultural practices that reduce environmental impacts while maintaining productivity and food security. Rice (Oryza sativa) and common bean (Phaseolus vulgaris) are staple crops, particularly in developing countries such as Brazil, playing a crucial role in human nutrition and food availability. However, conventional production systems rely heavily on synthetic inputs, which are associated with environmental degradation, soil health decline, and potential risks to human health.

Bioinputs, including biofertilizers, biopesticides, and biostimulants, have emerged as promising alternatives, offering potential benefits such as improved soil quality, reduced chemical dependency, and enhanced sustainability. Despite the increasing number of studies on bioinputs, the available evidence remains fragmented, with variability in methodologies, crops, environmental conditions, and reported outcomes. Additionally, few studies explicitly integrate agronomic performance with environmental and health-related dimensions under a planetary health perspective.

Therefore, this systematic review aims to synthesize and critically evaluate the existing evidence on the use of bioinputs in rice and bean production systems, providing a comprehensive and updated understanding of their agronomic, environmental, and health implications. This review seeks to address current knowledge gaps, reduce inconsistencies in the literature, and support evidence-based decision-making in sustainable agriculture and food systems.

Review objectives

To identify the types of bioinputs used in rice and bean production, including their functional categories (e.g., biofertilizers, biopesticides, and biostimulants);

To assess the agronomic impacts of bioinputs on crop productivity and soil health;

To investigate the role of bioinputs in reducing dependency on synthetic inputs and their associated environmental benefits;

To analyze how the adoption of bioinputs aligns with planetary health principles, considering implications for human health and sustainable food systems.

Keywords

Agriculture; Sustainability; Environmental impact; Food security; Public health

Country

Brazil

ELIGIBILITY CRITERIA

Population

Included

Studies will be included if they evaluate rice (Oryza sativa) and/or common bean (Phaseolus vulgaris) production systems under agricultural conditions. The population of interest comprises crop-based production systems, including field trials, greenhouse experiments, and controlled studies, in which these crops are cultivated and evaluated.

There will be no restrictions regarding geographic location, climatic conditions, soil type, or production system, provided that the study presents results applicable to agricultural contexts. Studies involving any growth stage of the crops will be considered, as long as agronomic, environmental, or sustainability-related outcomes are reported.

Excluded
Studies will be excluded if they do not involve rice (Oryza sativa) and/or common bean (Phaseolus vulgaris) production systems. Studies focusing on other crops without direct applicability to these species will not be considered.

Additionally, studies conducted exclusively under laboratory conditions without clear relevance to agricultural systems, as well as studies lacking agronomic, environmental, or sustainability-related outcomes, will be excluded. Studies that do not clearly describe the production system or crop context will also be excluded.

Intervention(s) or exposure(s)

Included
Biological Treatment

Eligible interventions will include the application of bioinputs of biological origin, such as biofertilizers, biopesticides, biostimulants, microbial inoculants, plant extracts, and algae-based products, applied in rice (Oryza sativa) and/or common bean (Phaseolus vulgaris) production systems. Bioinputs may be applied via soil, foliar spraying, seed treatment, or other agronomic practices, provided that the intervention is clearly described.

Studies will be included if they report measurable agronomic, environmental, or sustainability-related outcomes associated with the use of bioinputs. No restrictions will be applied regarding dose, frequency, formulation, or stage of crop development, as long as the intervention is explicitly linked to the evaluated outcomes.

Excluded
Studies will be excluded if they do not involve the application of bioinputs of biological origin in rice (Oryza sativa) and/or common bean (Phaseolus vulgaris) production systems. Interventions based exclusively on synthetic chemical inputs, such as mineral fertilizers or chemical pesticides, will not be considered unless used as a comparison treatment.

Additionally, studies in which the intervention is not clearly described, lacks sufficient methodological detail, or does not allow the identification of the type, mode of application, or function of the bioinput will be excluded. Studies that do not report measurable agronomic, environmental, or sustainability-related outcomes associated with the intervention will also be excluded.

Comparator(s) or control(s)

Included
PICO tags selected: Biological Treatment

Comparators may include conventional agricultural practices, such as the use of synthetic fertilizers and chemical pesticides, untreated control groups, or alternative management strategies. Studies comparing different types or formulations of bioinputs will also be considered.

No restrictions will be applied regarding the type of comparator, provided that the comparison allows for the assessment of agronomic, environmental, or sustainability-related outcomes. Studies without a comparator group will also be included when they provide relevant and measurable outcomes associated with the intervention.

Excluded

Studies will be excluded if the comparator is not clearly defined or does not allow meaningful interpretation of the results. Comparators unrelated to agricultural practices, such as laboratory-only controls without agronomic relevance, will not be considered.

Additionally, studies in which the comparator does not enable the evaluation of agronomic, environmental, or sustainability-related outcomes, or where the comparison is not adequately described, will be excluded.

Study design

Both randomized and nonrandomized study types will be included.

Included

This review will include experimental studies conducted under field, greenhouse, or controlled conditions, including randomized and non-randomized trials evaluating the use of bioinputs in rice (Oryza sativa) and/or common bean (Phaseolus vulgaris).

Systematic reviews and meta-analyses addressing similar topics will also be considered to support evidence synthesis and contextual discussion. Observational studies may be included if they provide relevant agronomic, environmental, or sustainability-related outcomes.

Excluded

The following types of studies will be excluded: opinion papers, narrative reviews, editorials, letters to the editor, conference abstracts, and publications without peer review. Studies lacking a clear methodological description or not providing measurable agronomic, environmental, or sustainability-related outcomes will also be excluded.

Additionally, purely laboratory-based studies without clear applicability to agricultural systems, as well as studies not related to rice (Oryza sativa) and/or common bean (Phaseolus vulgaris), will not be considered.

Context

This review will consider studies conducted in agricultural production systems of rice (Oryza sativa) and/or common bean (Phaseolus vulgaris) under field, greenhouse, or controlled conditions. There will be no restrictions regarding geographic location, climatic conditions, soil types, or production systems (e.g., conventional, organic, or agroecological), provided that the studies are relevant to agricultural practices.

The context includes diverse edaphoclimatic conditions and management systems, aiming to capture variability in the performance of bioinputs across different production environments and their implications for sustainability and food systems.

TIMELINE OF THE REVIEW

Date of first submission to PROSPERO

10 April 2026

Review timeline

Start date: 4 August 2025. End date: 5 October 2026.

Date of registration in PROSPERO

13 April 2026

AVAILABILITY OF FULL PROTOCOL

Availability of full protocol
A full protocol has been written and uploaded to PROSPERO. The protocol will be made available after the review is completed.

SEARCHING AND SCREENING

Search for unpublished studies
Only published studies will be sought.

Main bibliographic databases that will be searched
The main databases to be searched are *PubMed* and *Scopus*.

Other important or specialist databases that will be searched
ScienceDirect

Web of Science

SciELO

Frontiers

Search language restrictions
There are no language restrictions.

Search date restrictions
Databases will be searched for articles published from 1 January 2015, there are no search end date restrictions.

Other methods of identifying studies
Other studies will be identified by: *contacting authors or experts*, *looking through all the articles that cite the papers included in the review ("snowballing" or forward citation searching)* and *searching trial or study registers*.

Additional information about identifying studies
Additional identification of studies may be conducted through manual screening of reference lists of included articles and relevant reviews. Citation tracking and complementary searches within the selected databases using alternative filters and search strategies may also be performed when necessary.

Link to search strategy
A full search strategy is available in the full protocol as described in the *Availability of full protocol* section

Selection process
Screening conducted using Rayyan platform.

Other relevant information about searching and screening
Search strategies will be refined iteratively, and screening will follow predefined criteria to ensure consistency and reproducibility.

DATA COLLECTION PROCESS

Data extraction from published articles and reports

Data extraction will be performed independently by reviewers using standardized forms, with discrepancies resolved by consensus.

Authors will be asked to provide any required data not available in published reports.

Study risk of bias or quality assessment

Risk of bias will be assessed using:

In addition to the JBI tools, methodological quality will be assessed based on predefined criteria, including clarity of objectives, adequacy of study design, appropriateness of statistical methods, transparency of reporting, and potential conflicts of interest.

Data will be assessed independently by at least two people (or person/machine combination) with a process to resolve differences.

Additional information will be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

Reporting bias assessment

Risk of bias due to missing results will be assessed

Certainty assessment

The certainty of evidence will be assessed through a critical appraisal of the methodological quality of the included studies, considering factors such as clarity of objectives, adequacy of study design, appropriateness of statistical methods, consistency of findings, and potential for generalization. Publication bias will also be qualitatively evaluated.

OUTCOMES TO BE ANALYSED

Main outcomes

Primary outcomes will include crop productivity (yield), soil health indicators, and the effectiveness of bioinputs in pest and disease control in rice (Oryza sativa) and common bean (Phaseolus vulgaris) production systems.

Additional outcomes

Additional outcomes will include resource use efficiency, reduction in synthetic input use, environmental sustainability indicators, and potential implications for human health and food systems.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

Data will be combined through a narrative synthesis, grouping studies according to crop, type and category of bioinput, application method, and reported agronomic and environmental outcomes. When appropriate, comparable data will be quantitatively analyzed to support the synthesis.

CURRENT REVIEW STAGE

Stage of the review at this submission

Review stage	Started	Completed
Pilot work	✓	✓
Formal searching/study identification	✓	✓
Screening search results against inclusion criteria	✓	
Data extraction or receipt of IPD		
Risk of bias/quality assessment		
Data synthesis		

Review status

The review is currently planned or ongoing.

Publication of review results

Results of the review will be published in English and Portuguese.

REVIEW AFFILIATION, FUNDING AND PEER REVIEW

Review team members

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

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Funding source

Review has no funding and no agreed support from an academic institution and is done in authors’ own time.

Peer review

The protocol was reviewed internally by the research team, including subject-matter experts, to ensure methodological rigor, clarity, and consistency prior to submission.

ADDITIONAL INFORMATION

Review conflict of interest

Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

Medical Subject Headings

Agricultural Inoculants; Biological Control Agents; Brazil; Food Security; Humans; Oryza; Outcome Assessment, Health Care; Phaseolus; Plant Extracts; Seeds; Soil

SIMILAR REVIEWS

Check for similar records already in PROSPERO

PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 10 April 2026). These are shown below along with the reasons given by that the review team for the reviews being different and/or proceeding.

- The Global Landscape of Rice Diversification: A Systematic Review of its Impacts on Crop Productivity, Income, and Nutrition [published 17 August 2025] [CRD420251113022]. The review was judged **not to be similar**
- Building Sustainable Workplaces: A Systematic Review of Green Human Resource Management (GHRM) Practices and Their Impact on Organisational Sustainability [published 12 November 2025] [CRD420251229461]. The review was judged **not to be similar**
- Biotechnological Assessment of Microplastic-Induced Gut Microbiome and Associated Gastrointestinal Health Outcomes in the Global human Population: A Systematic Review and Meta-Analysis [published 11 March 2026] [CRD420261302781]. The review was judged **not to be similar**
- Developing Environmental Sustainability course in health-related Programs [published 10 September 2024] [CRD42024567001]. The review was judged **not to be similar**
- Life Cycle Assessment of Hydrogen Fuel Cell Buses: A Systematic Review of Methodological Approaches [published 29 December 2025] [CRD420251275635]. The review was judged **not to be similar**
- Advancing Sustainable Child Nutrition: Integrated Strategies and Growth Outcomes — A Systematic Review and Meta-Analysis [published 13 October 2025] [CRD420251168138]. The review was judged **not to be similar**
- Environmental and Economic Implications of Anesthesia Practice in the Operating Room: A Systematic Review of Gas Selection, Flow Strategies, and Nitrous Oxide Use [published 24 January 2026] [CRD420251244190]. The review was judged **not to be similar**
- Ultrasound goes echo: Decarbonizing Inflammatory bowel disease (IBD) Care through Intestinal Ultrasound – A Systematic Review [published 2 August 2025] [CRD420251088016]. The review was judged **not to be similar**
- A Systematic Review of Water Pollution, Key Pollutants, and Environmental and Health Implications in Bangladesh [published 24 November 2024] [CRD42024614202]. The review

was judged **not to be similar**
- Environmental and health impact of school feeding programmes: A systematic review [published 18 August 2024] [CRD42024559619]. The review was judged **not to be similar**
- Incentives and barriers affecting the transition towards the circular economy: a systematic literature review [published 3 March 2024] [CRD42024516375]. The review was judged **not to be similar**
- Structural Determinants and Therapeutic Interventions in Pediatric Respiratory Morbidity: A PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)-Guided Systematic Review and Random-Effects Meta-Analysis with Cost-Effectiveness Evaluation [published 7 March 2026] [CRD420261325701]. The review was judged **not to be similar**
- Architectural, structural and urban health promotion interventions in school settings and their impact on health-related outcomes of the school community: a systematic review [published 21 January 2026] [CRD420261282226]. The review was judged **not to be similar**
- Impact of Aflatoxins on Human Health Outcomes: A Systematic Review [published 16 July 2025] [CRD420251077128]. The review was judged **not to be similar**
- Education for Sustainable Development Training Programs for Physical Education Teachers: A Systematic Review [published 4 December 2025] [CRD420251247056]. The review was judged **not to be similar**
- Plastics in the food system: impacts on human health, environment and economic sustainability outcomes. A systematic review [published 25 October 2018] [CRD42018110276]. The review was judged **not to be similar**
- What is the Real Impact of Skipping Breakfast on Overall Health? [published 23 March 2025] [CRD420251017983]. The review was judged **not to be similar**
- Agriculture, food and nutrition security interventions that facilitate sustainable development and have a positive impact on health: an overview of systematic reviews [published 12 March 2014] [CRD42014008780]. The review was judged **not to be similar**
- Improving universal access to planetary health diets, within planetary boundaries, while advancing justice: A systematic review of effective interventions across food systems [published 8 May 2023] [CRD42023420048]. The review was judged **not to be similar**
- Effectiveness of Agricultural Extension Programs in Promoting Safe Wastewater Reuse for Irrigation and Reducing Food-Borne Health Risks: A Systematic Review Protocol. [published 13 November 2025] [CRD420251208373]. The review was judged **not to be similar**

PROSPERO version history

- Version 1.0, published 13 Apr 2026

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